

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler with experience developing and validating predictive models for financial-risk monitoring, commercial property insurance pricing, and large dependent datasets. Combines GLM/GBM, econometrics, Monte Carlo simulation, optimization, and statistical diagnostics with reproducible Python/PyTorch workflows, translating business and risk questions into independently developed, testable models.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Data: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; data cleaning, feature processing, missingness handling, and CPU/GPU computing

Modeling and Analytics: Predictive modeling, GLM/GBM, econometrics, machine learning, MLE/GLS, Monte Carlo simulation, numerical optimization, hypothesis testing, stochastic processes, model selection, validation, and performance diagnostics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a business monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and Monte Carlo simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ commercial property-risk records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and diagnostics on AWS EC2.
- Evaluated predictive performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE MODELING AND RESEARCH

Ph.D. Dissertation Research - Scalable Model Development and Diagnostics

Sep 2024 - Present

- **Independent Model Development:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- **Data Processing and Cleaning:** Built workflows for quality filtering, missingness tracking, time-dependent coordinate offsets, nearest-neighbor center matching, and regular-grid construction to correct systematic measurement shifts.
- **Model Evaluation:** Designed simulation studies and frequency-resolved diagnostics to identify whether misspecification arises from low-frequency structure, high-frequency behavior, noise, or dependence.
- **Scalable Research Engineering:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs; integrated C++ routines through pybind11.

Econometrics Project - Earned Income Tax Credit and Household Labor Supply

Graduate Research

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household covariates.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra

Languages: Korean (Native); English (Fluent); Chinese (Basic)